

Predicting the Right Treatment for the Right Patient: An AI-Powered Decision Support Framework Based on PITE

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Motivation

Medical decisions:

diagnosis and prognosis treatment selection

are inherently subject to uncertainty

- Not all participants will react equally to an intervention
- Characterizing this heterogeneity in intervention effects is key to improving patients outcomes

The predicted individual treatment effects (PITE) framework

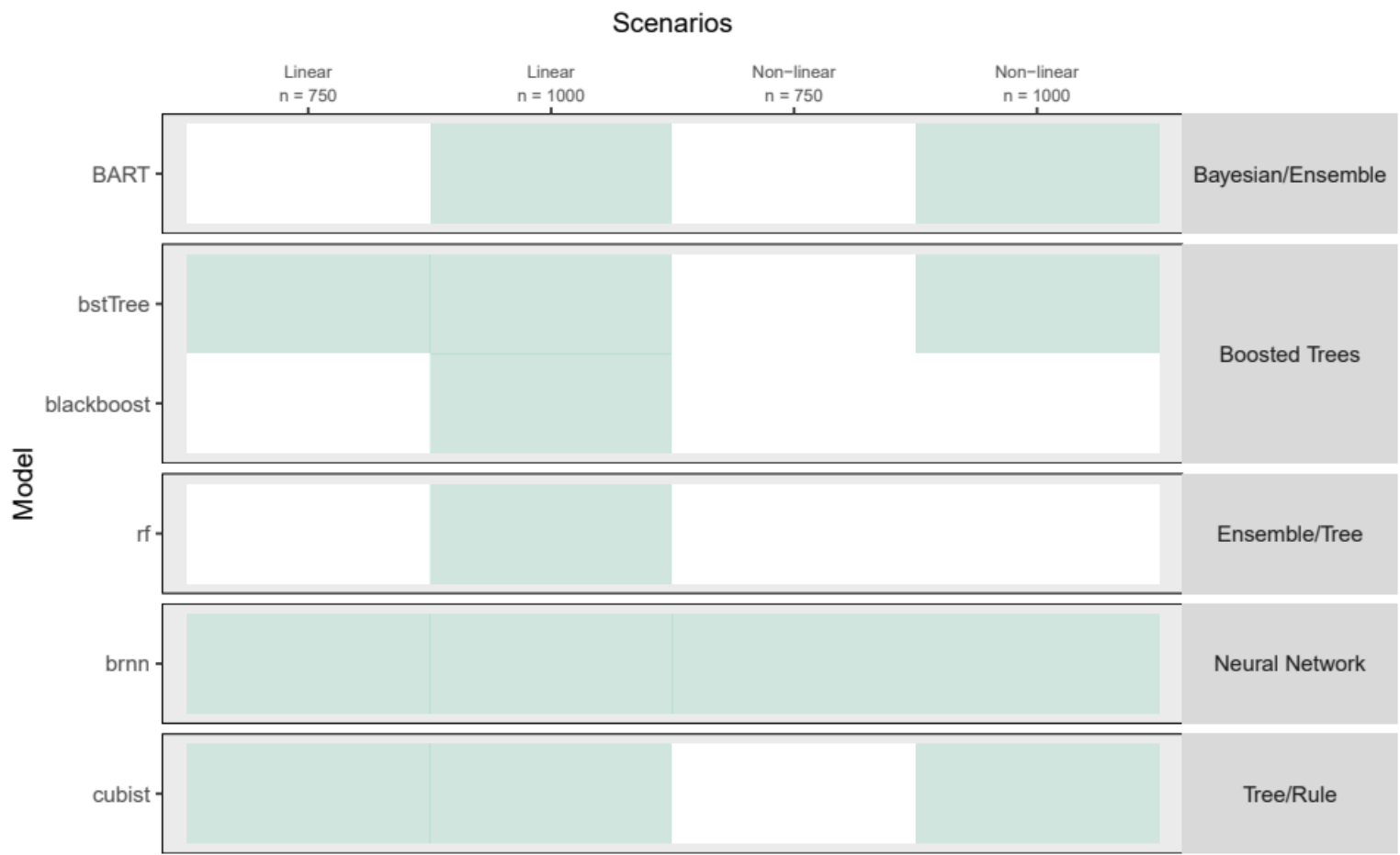
- PITE consists of the difference between experimental(E) and control (C) **prediction** for each individual (Jaki et al. 2024).

$$\text{PITE}_i = f_E(X_i) - f_C(X_i), \quad f(\cdot) \text{ is a predictor}$$

! Challenge

- PITE is unobserved.
- The best method to estimate the PITE is not necessarily the best fitted model.

Modeling Strategies



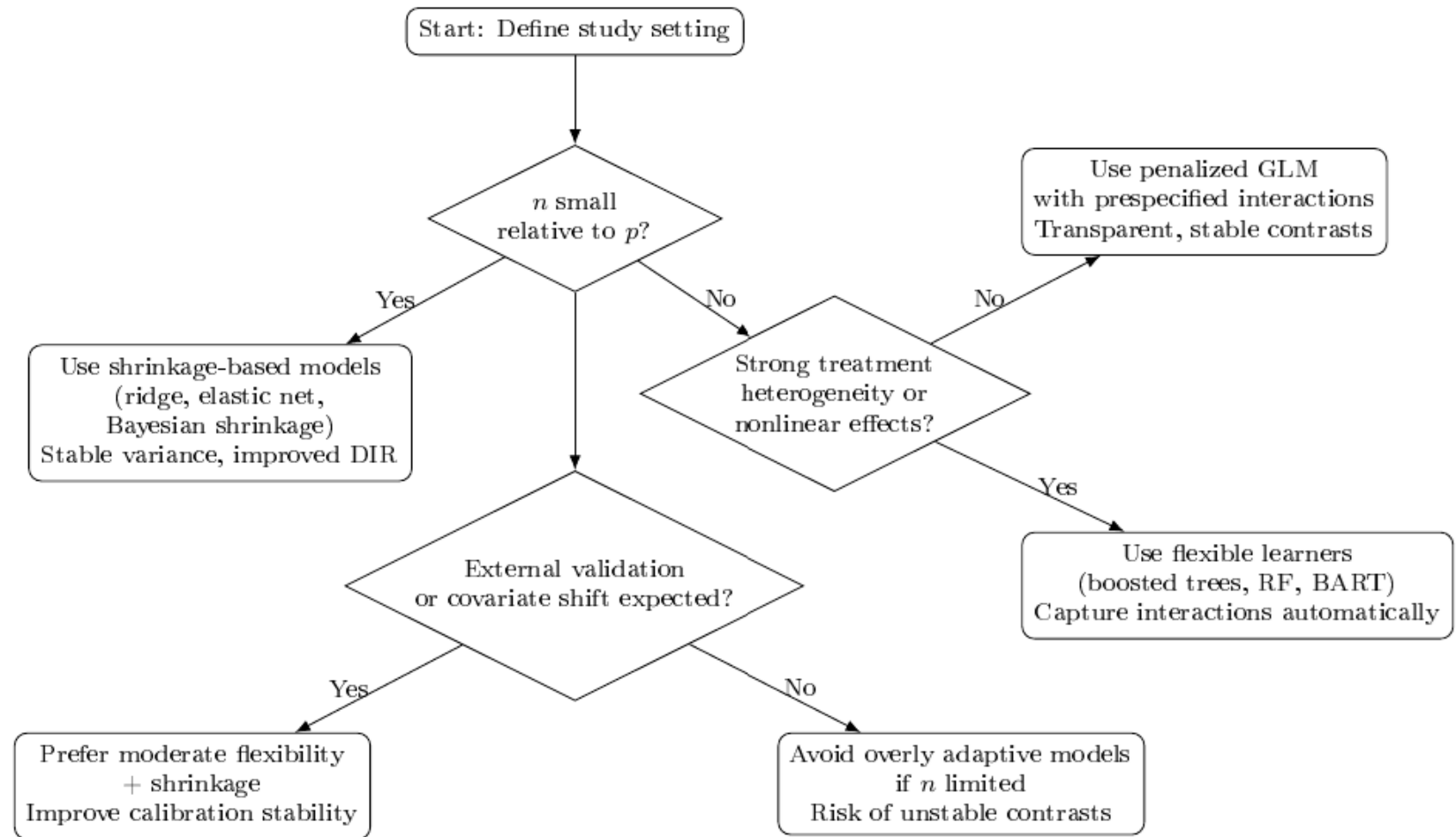
External validation scenarios with good PITE performance

RMSE <1.5 and correct recommendation > 0.65.

Moderate Heterogeneity.

Solano et. al (2026)

Bayesian Learning Strategies for PITE

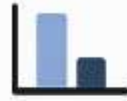


Solano et. al (2026)

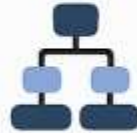
Figure 10: Decision framework for robust estimation of predicted individual treatment effects (PITEs). The roadmap summarizes simulation-based evidence on variance control, heterogeneity capture, and transportability.

Complex outcome

Solano et. al (2026 b)



Tailored to AUD outcomes with mass at 0 and 1



Interpretable: Matches clinical decision structure (abstinence, partial, persistent use)



Powerful: Shared tree ensemble improves predictions and stability



Scalable: Fast conjugate-friendly updates via Beta approximation



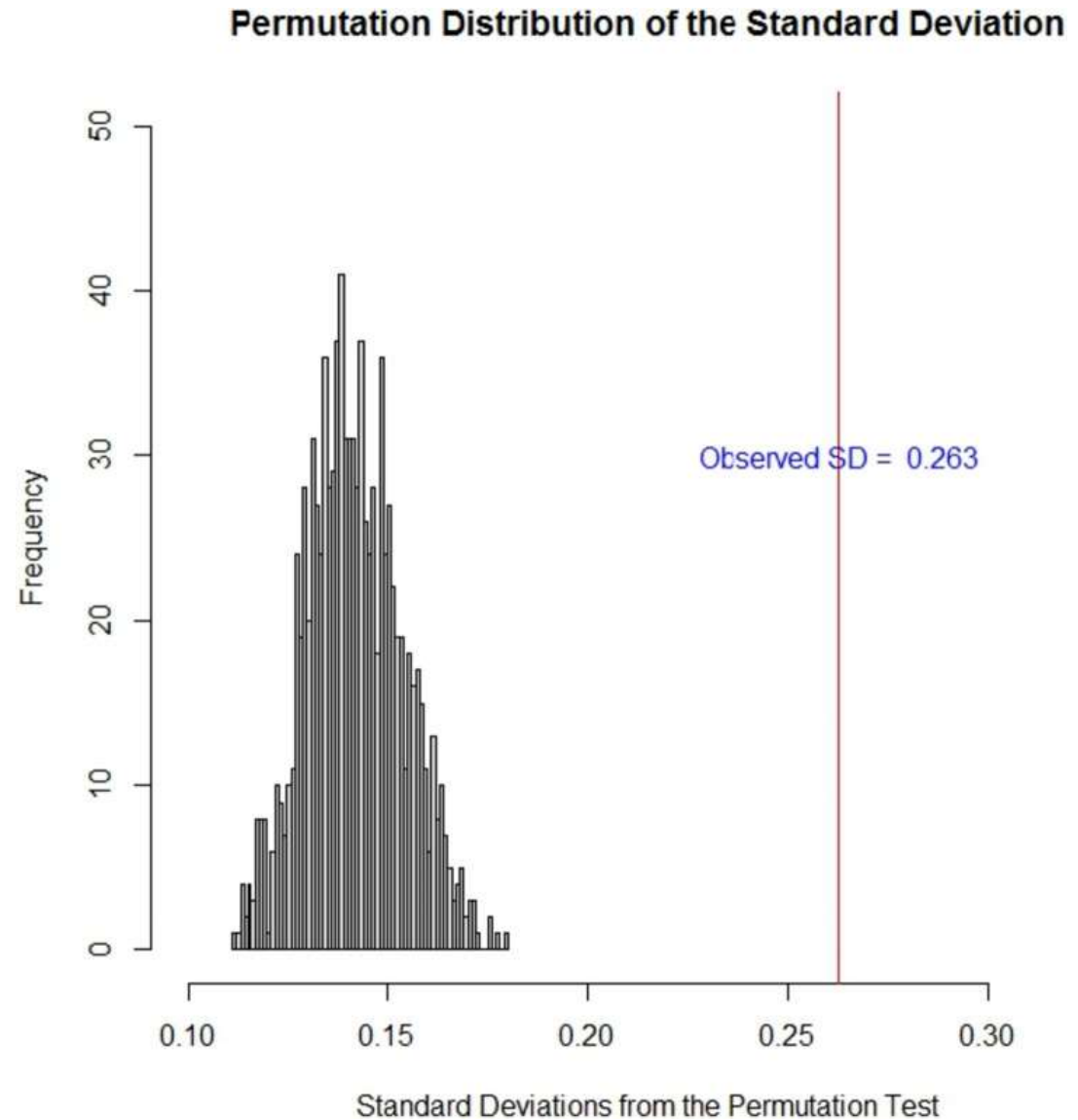
Actionable: Enables individual-level treatment effect (PITE) estimation



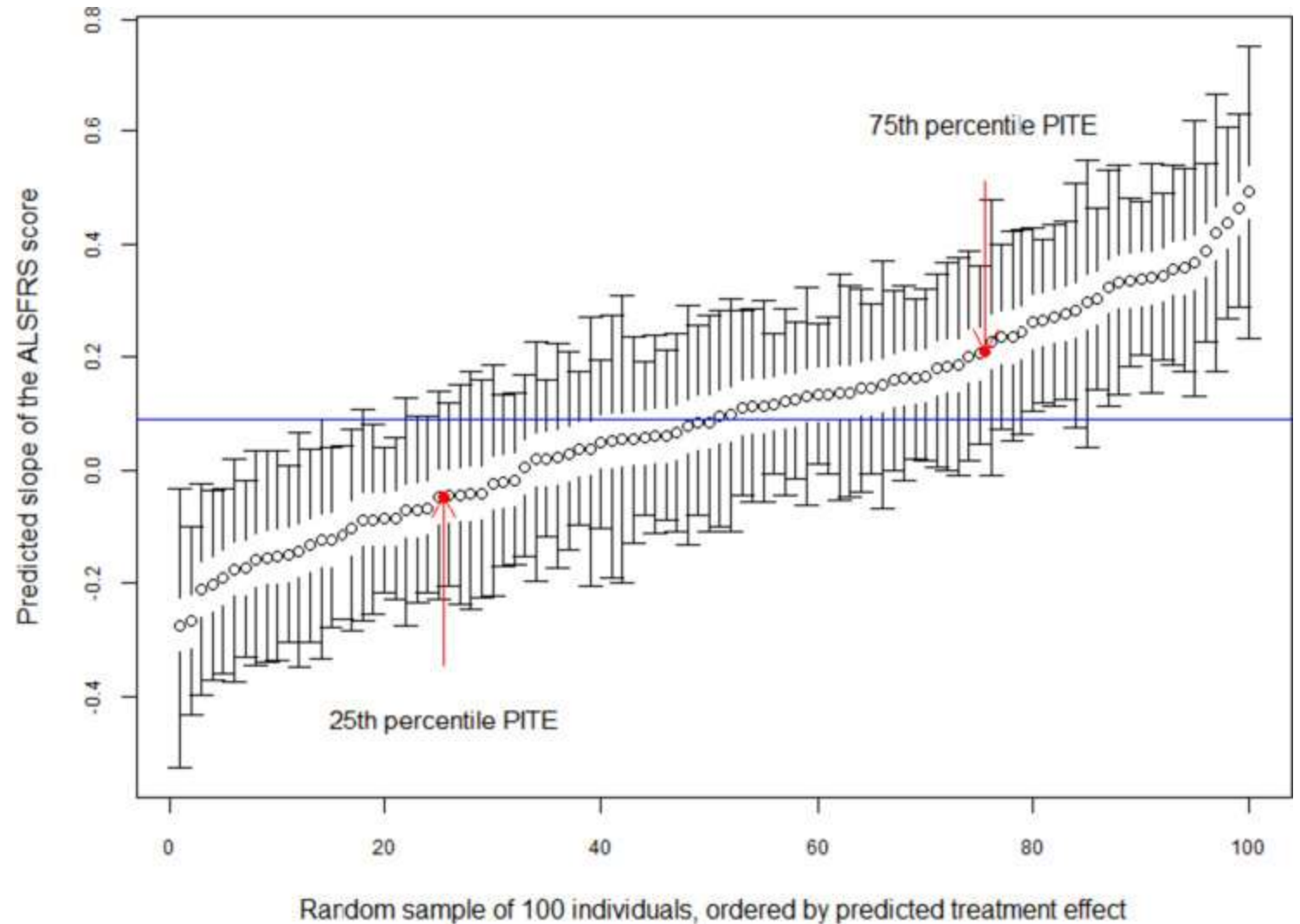
Proven: Outperforms ZOIB in both simulation and real clinical data

Complex outcome

Jaki et. al (2025)



Complex outcome



Jaki et. al (2025)

Missing Information and PITE

$$\tau(X_i) = f_1(X_i) - f_0(X_i)$$

- PITE depends on the covariate information available for each patient.

Under an appropriate imputation model [Rubin, 1987]

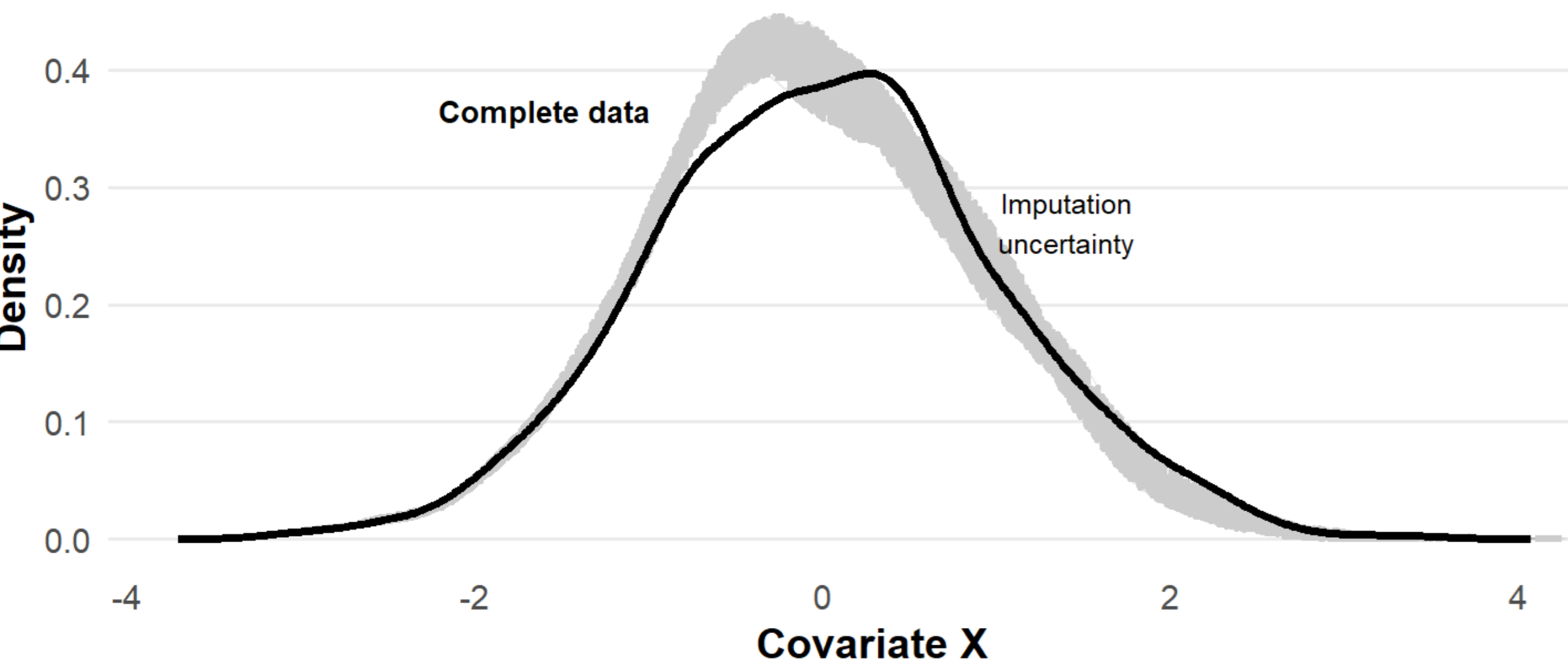
- MI is specifically designed to propagate missing-data uncertainty.

Imputation's vulnerability [Sterne et al., 2009]

- Based on assumptions that do not hold.
- Imputation model can be misspecified

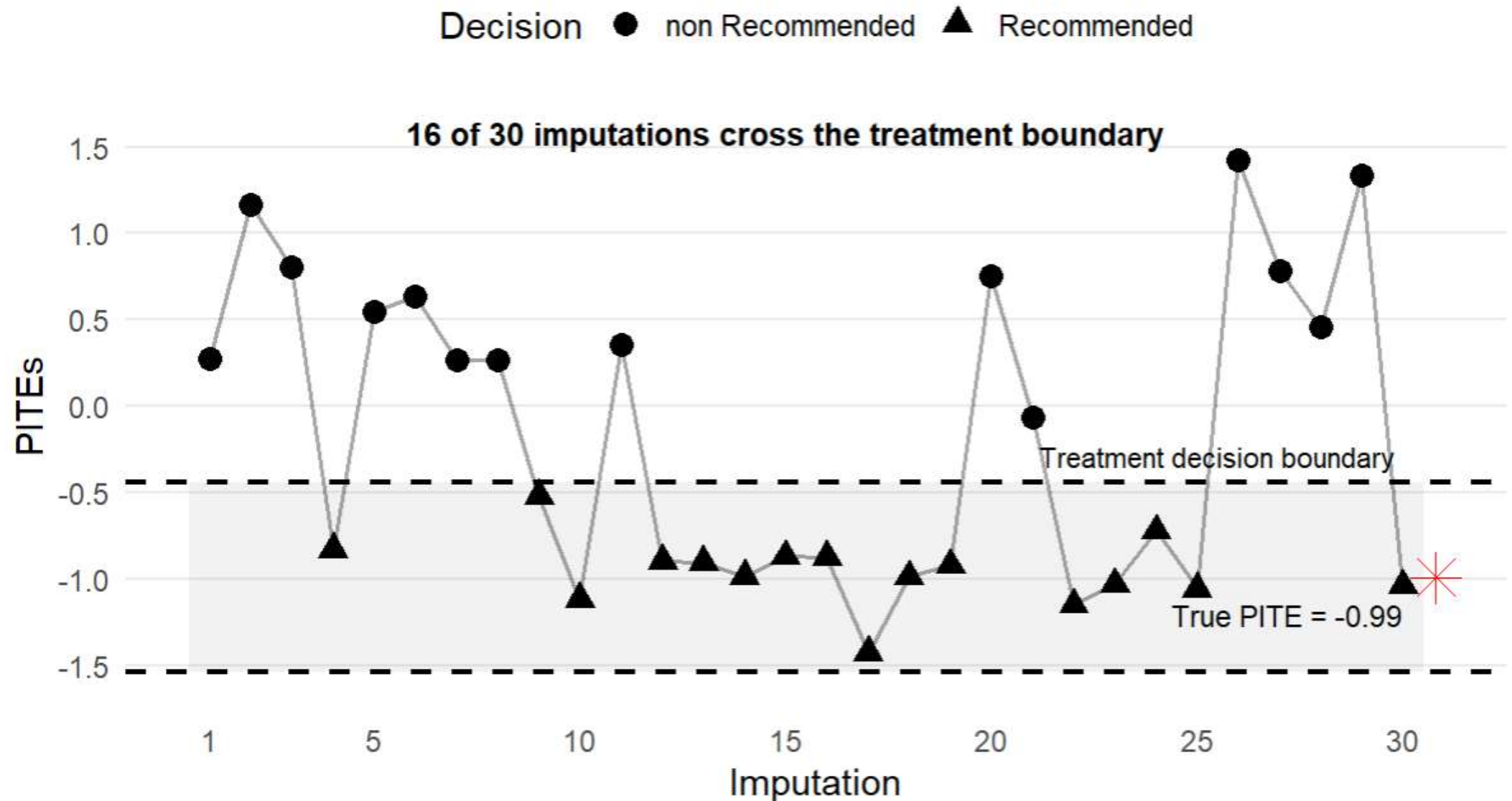
Imputation changes the predictor distribution learned by PITE

Black = complete data | Grey = 30 imputed datasets | band = 10th–90th percentile across imputations



Same patient. Different imputation. Different treatment decision

Patient 15 | Mean PITE = -0.21 | SD = 0.87



Conclusion

- Clinical trials traditionally focus on average treatment effects
- PITE provides an individualized perspective
- Machine-learning methods such as BART can capture
 - nonlinear relationships and
 - interactions that may be missed by simpler models.
- Challenges: PITE validation, choice of prediction method, and translating predictions into reliable treatment decisions.
- PITE can be affected by missing patient information:
- Problem: The reliability and stability of PITEs

Thank you

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